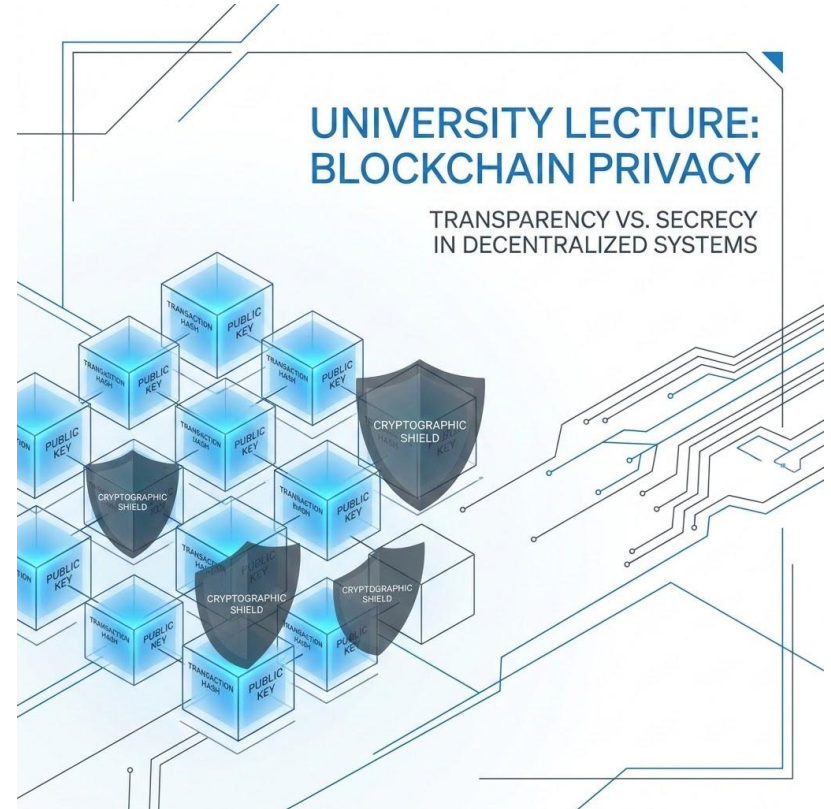


Blockchain and Cryptocurrencies Technologies

Privacy and Anonymity in Cryptocurrencies

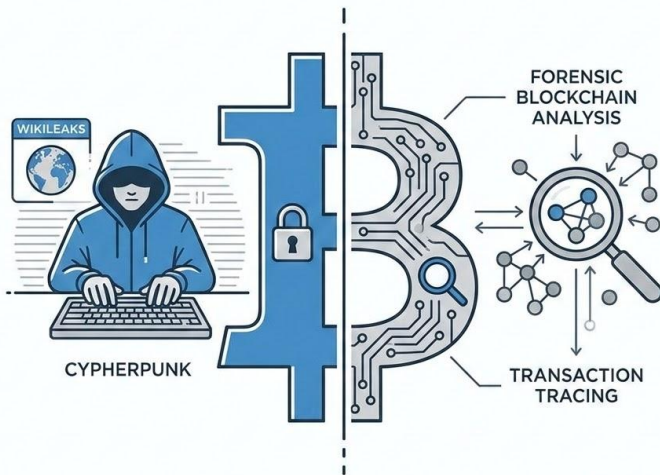
Privacy and Anonymity in Cryptocurrencies

- Blockchain and Cryptocurrencies Technologies
- From public ledgers to privacy-preserving systems
- Why privacy is hard in transparent financial infrastructures
- How cryptographic design changes what can and cannot be traced



Is Bitcoin Anonymous?

IS BITCOIN ANONYMOUS?



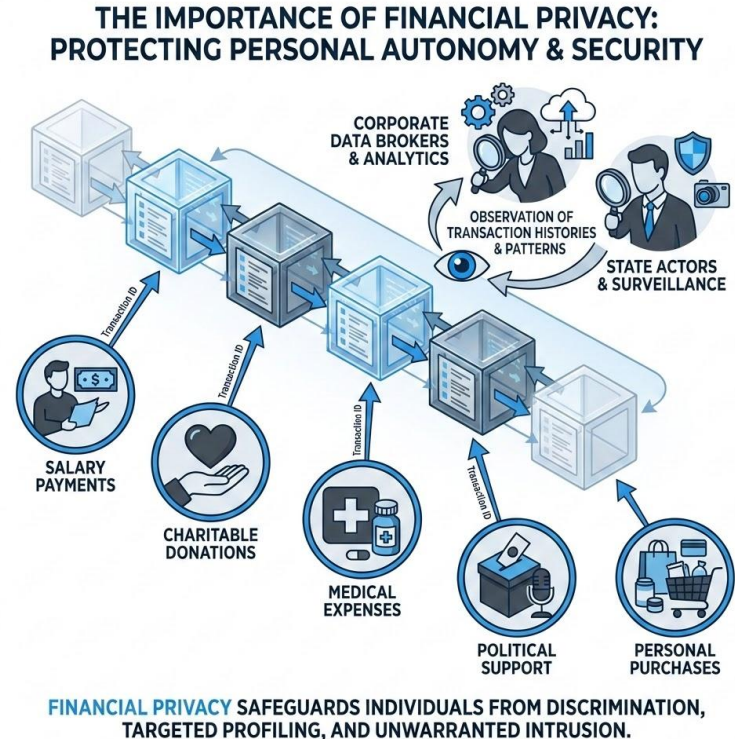
2011: CYPHERPUNK VISION

MODERN FORENSICS

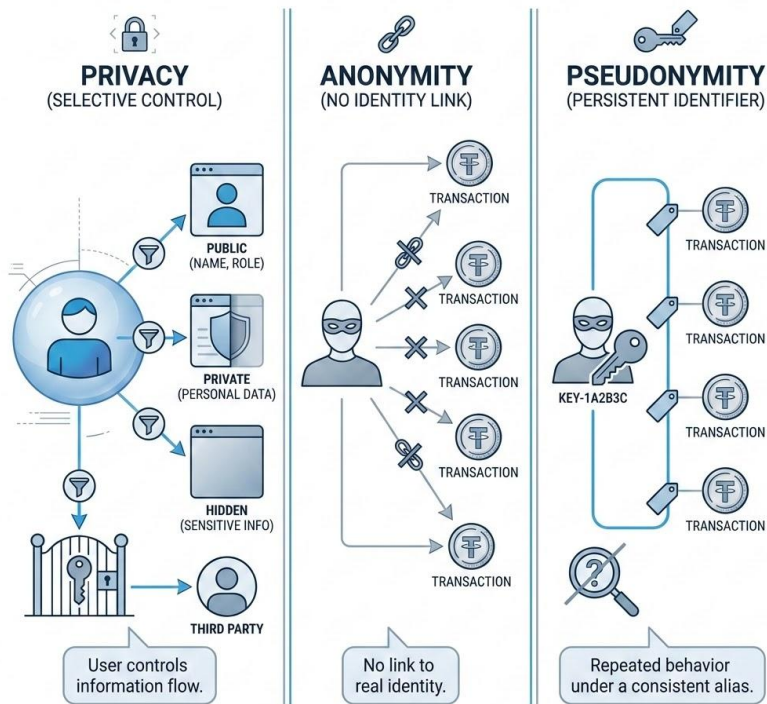
- Early narratives described Bitcoin as “anonymous” because it removed names from transactions
- Law enforcement and chain analytics firms argue the opposite: Bitcoin is highly traceable
- Both views capture part of the truth, depending on how privacy is defined
- This lecture starts from that tension:
 - Anonymity
 - Pseudonymity
 - Unlinkability
 - untraceability
- The key question is not whether blockchain uses names, but whether behavior can still be linked to people

Why Privacy Matters in Financial Systems

- Financial privacy protects more than crime: it protects ordinary life
- Public exposure of payments can reveal:
 - Salaries
 - Donations
 - medical spending
 - political preferences
 - personal relationships
- Transparent ledgers can enable:
 - surveillance by firms
 - surveillance by governments
 - targeted coercion
 - deanonymization at scale
- Privacy is therefore not a luxury feature, but a structural property of freedom, autonomy, and safety



Privacy Is Not the Same as Anonymity

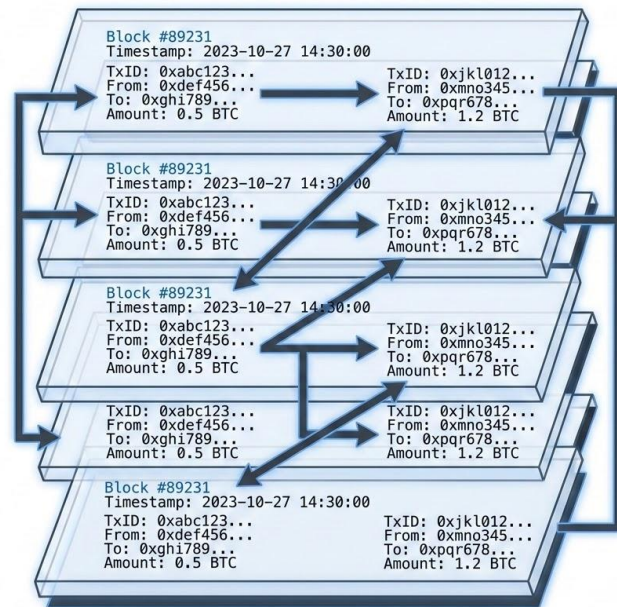


- Privacy = control over what others can learn about you
- Anonymity = actions cannot be directly associated with your identity
- Pseudonymity = actions are attached to a stable identifier that is not your legal name
- Public-key blockchains typically provide pseudonymity, not anonymity
- Once a pseudonym is linked to a person:
 - past activity becomes visible
 - future activity becomes easier to monitor
- The distinction matters because many blockchain systems remove names but preserve behavioral continuity

Transparency Is a Feature, and a Privacy Problem

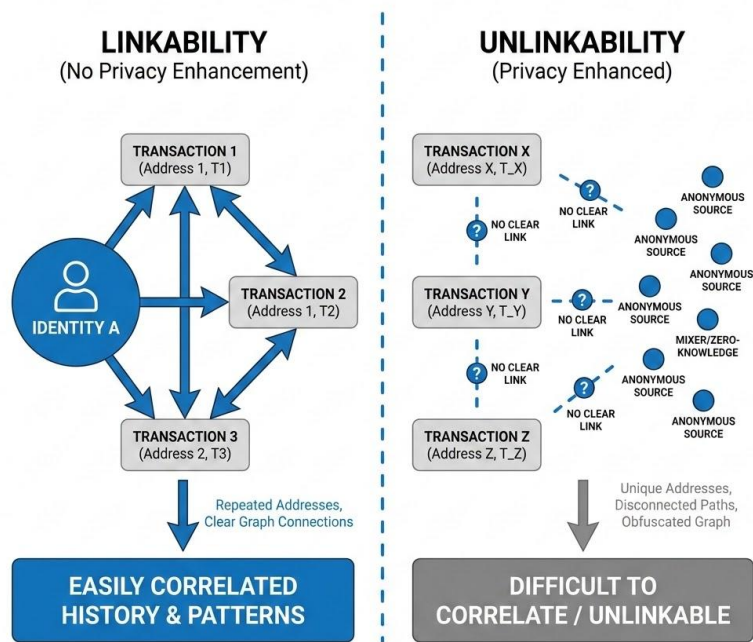
- Public blockchains make transaction history globally visible
- Even without legal names, the ledger exposes:
 - transaction graph structure
 - Amounts
 - Timing
 - repeated behavioral patterns
- Privacy is weakened because:
 - data is persistent
 - data is searchable
 - data is globally replicable
- In a public ledger, deleting identifiers does not eliminate inference
- This is why blockchain privacy is fundamentally harder than “just not storing names”

PUBLIC BLOCKCHAIN LEDGER: VISIBLE TRANSACTIONS & TRACEABLE PATTERNS



Note: All transactions, timestamps, and amounts are publicly visible. Identity is pseudonymized (no names), yet patterns remain traceable.

Unlinkability as a Stronger Privacy Goal



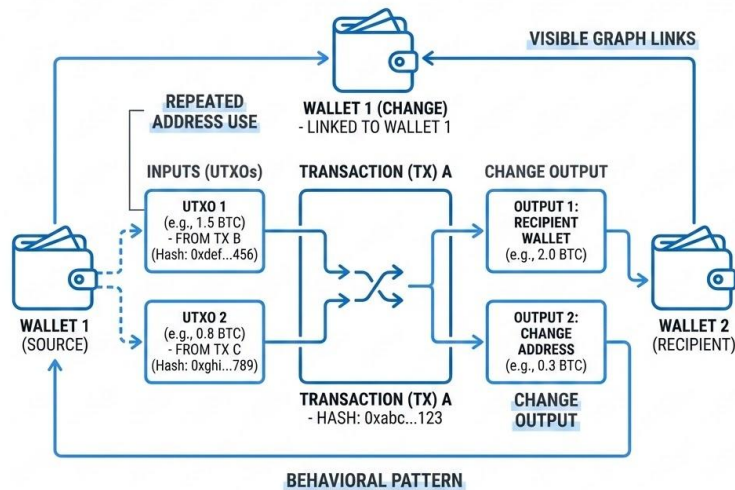
ACADEMIC DIAGRAM ILLUSTRATING TRANSACTION LINKABILITY AND UNLINKABILITY

- A private system should not only hide names; it should make actions hard to link together
- Unlinkability means:
 - different transactions cannot be confidently attributed to the same actor
- In practice, unlinkability is attacked through:
 - repeated address use
 - transaction graph analysis
 - timing correlation
 - external information sources
- The stronger the linkage between actions, the weaker the privacy of the system
- In blockchain privacy research, unlinkability is often more useful than the simplistic question “is it anonymous?”

Why Bitcoin Is Pseudonymous, Not Anonymous

- Bitcoin transactions are public, permanent, and graph-structured
- Wallets are identifiers, even if they are not legal identities
- Reusing addresses or combining inputs creates visible patterns
- The UTXO model helps validation, but also leaks structure:
 - input aggregation
 - change outputs
 - spending behavior
- This makes Bitcoin traceable through behavior, not just through explicit identifiers
- As a result, Bitcoin is better described as pseudonymous and partially linkable

BITCOIN TRANSACTION STRUCTURE: UTXOs, INPUTS, OUTPUTS & VISIBLE WALLET GRAPH

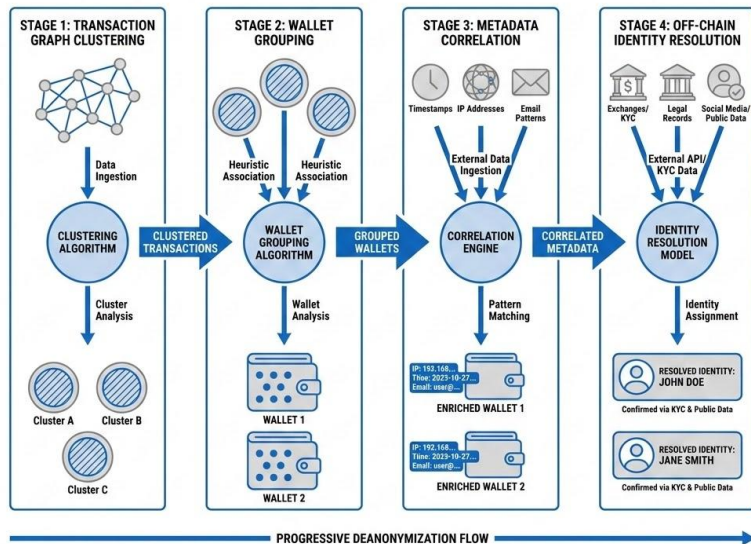


NOTE: WALLET GRAPH REVEALS PATTERNS THROUGH UTXO FLOWS AND CHANGE ADDRESSES, WITHOUT PERSONAL NAMES.

How Deanonymization Actually Works

BLOCKCHAIN DEANONYMIZATION PIPELINE: A FOUR-STAGE ACADEMIC ANALYSIS

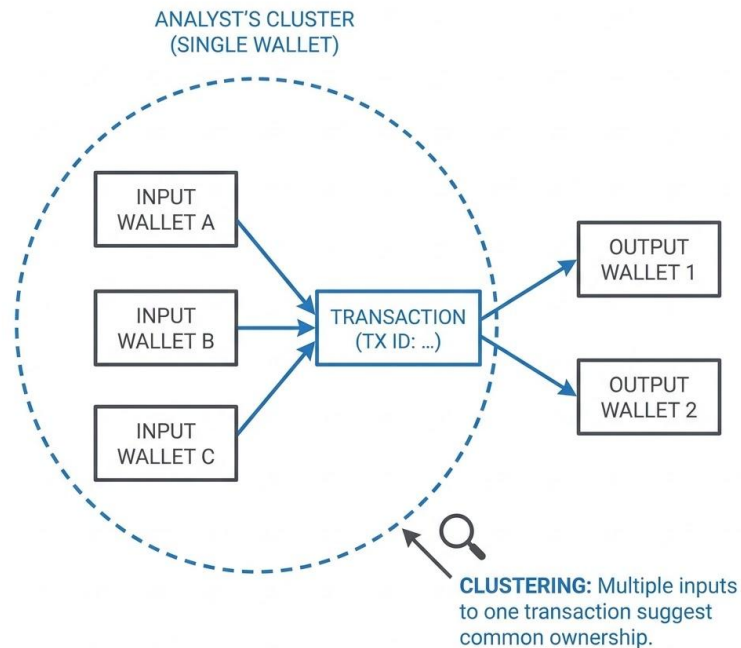
BLOCKCHAIN DEANONYMIZATION PIPELINE: A Four-Stage Academic Analysis



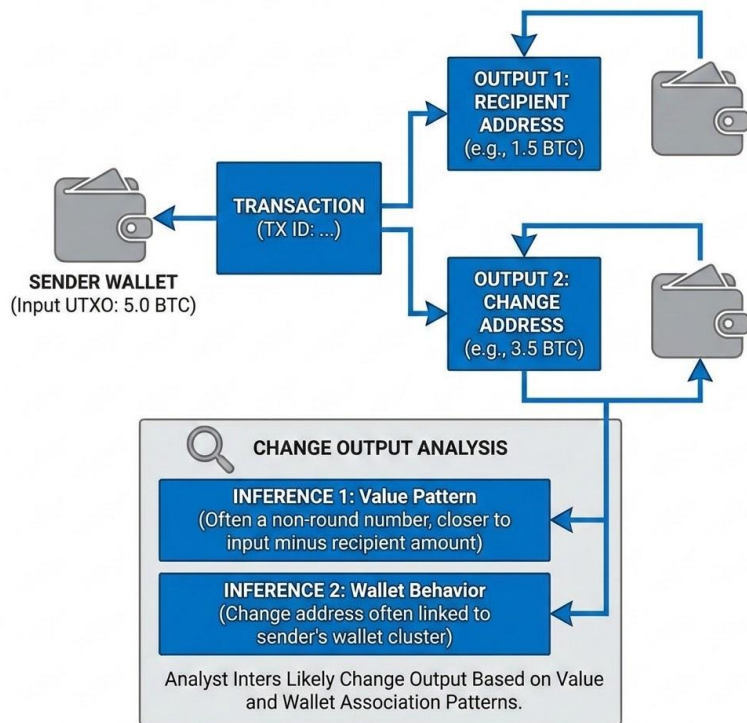
- Deanonymization is rarely a single attack; it is a pipeline
- Typical stages:
 - graph clustering
 - wallet grouping
 - metadata correlation
 - off-chain identity resolution
- Blockchain analysis firms combine:
 - transaction structure
 - Timing
 - IP-level traces
 - exchange records
 - merchant and service data
- Privacy fails when on-chain patterns and off-chain records are joined together
- The key lesson:
 - the ledger alone may not identify you
 - but the ledger plus the real world often does

Multi-Input Transactions and Wallet Clustering

- In Bitcoin, a transaction may use multiple inputs to assemble enough value
- A common heuristic assumes:
 - if several inputs are signed together, they are likely controlled by the same entity
- This is called the multi-input heuristic
- It is not logically perfect, but often operationally powerful
- The heuristic enables clustering of multiple addresses into one likely owner group
- Thus, transaction efficiency can directly reduce privacy



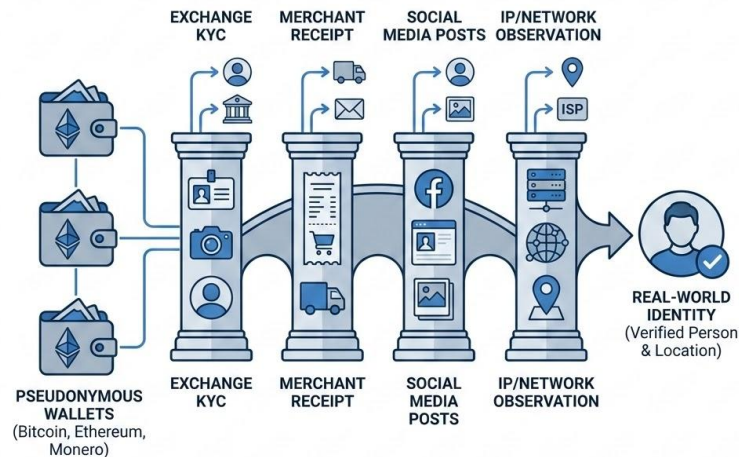
Change Outputs Leak Ownership Patterns



- Bitcoin usually spends an entire UTXO, then returns the unused remainder as change
- The recipient receives one output; the sender receives another output back
- Analysts infer which output is change using clues such as:
 - value structure
 - address reuse patterns
 - script types
 - wallet behavior
- Once change is identified, the sender's wallet cluster becomes easier to expand
- Change detection is one of the most important privacy leaks in Bitcoin's transaction model

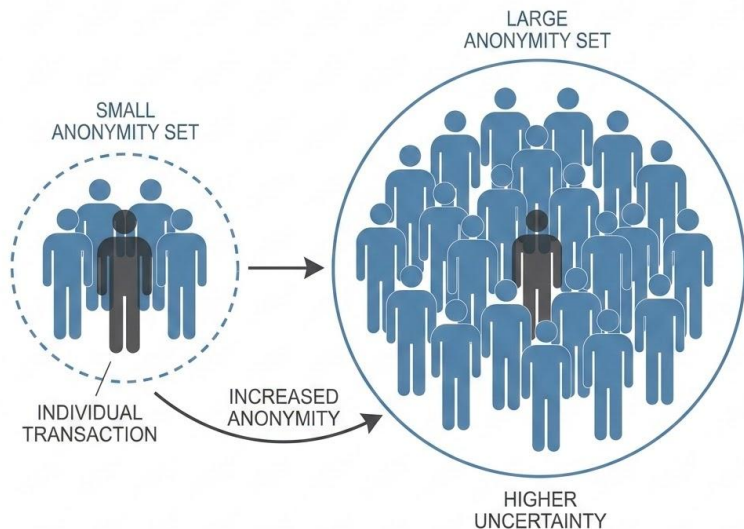
The Real World Destroys Blockchain Privacy

- On-chain pseudonyms often become identifiable through off-chain events
- Common linking points include:
 - exchange KYC
 - posted donation addresses
 - merchant data
 - invoices and receipts
 - IP-level observation
 - reused public profiles
- Once a wallet is linked to a person, past and future activity can often be inferred
- Public ledgers amplify the impact of one identity leak into a long-term surveillance trail
- This is why privacy in blockchain cannot be solved purely at the wallet layer



DEANONYMIZATION PATHWAYS: LINKING WALLETS TO IDENTITY THROUGH EXTERNAL DATA POINTS

Quantifying Privacy Through the Anonymity Set

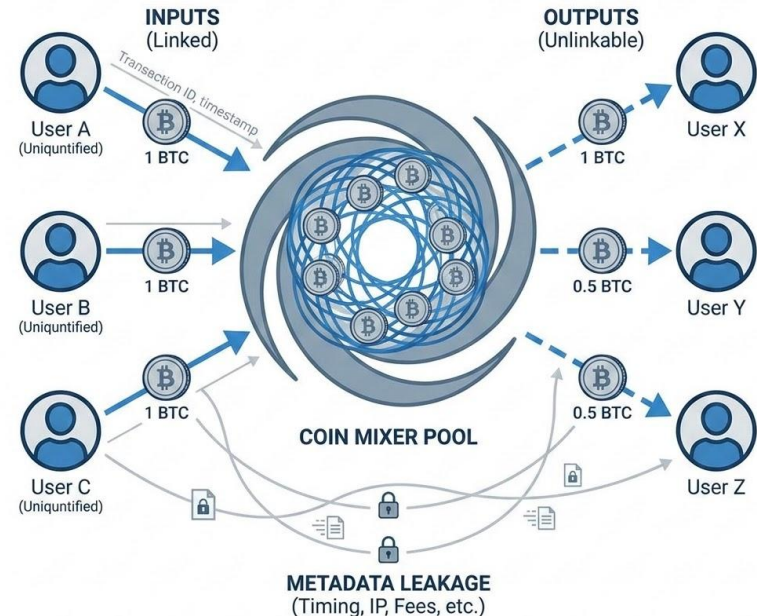


- Privacy is often measured by the size of the crowd you can hide inside
- The anonymity set = the number of plausible actors a transaction could belong to
- A larger anonymity set usually means better privacy
- But size alone is not enough:
 - bad timing
 - side channels
 - off-chain leakage
 - can still collapse privacy
- Real anonymity is contextual, dynamic, and often fragile
- Good privacy systems aim to maximize uncertainty for the observer

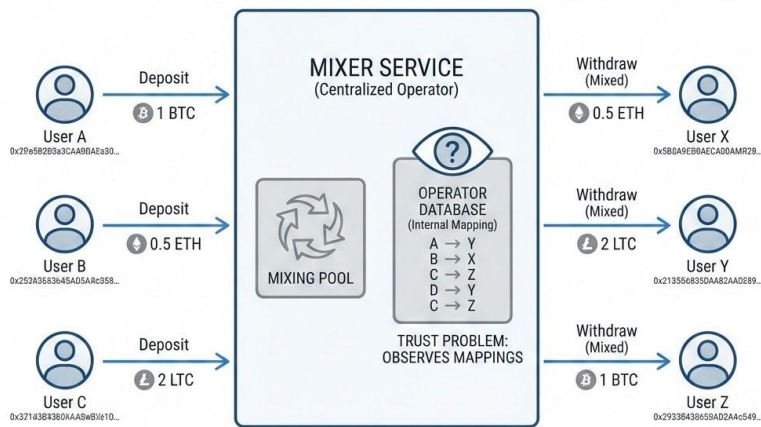
Mixing as a Response to Public Traceability

- If the ledger is transparent, one way to regain privacy is to break input-output linkage
- Mixers aim to obscure the path from sender to recipient by reshuffling funds
- Core intuition:
 - many users enter a pool
 - users exit in a way that obscures origin
- This is a workaround on top of a transparent ledger, not a redesign of the ledger itself
- Mixing improves ambiguity, but often leaves:
 - metadata leakage
 - trust assumptions
 - coordination challenges

CONCEPTUAL COIN MIXER DIAGRAM: BROKEN LINKAGE & METADATA RISKS



Centralized Mixers: Useful but Fragile

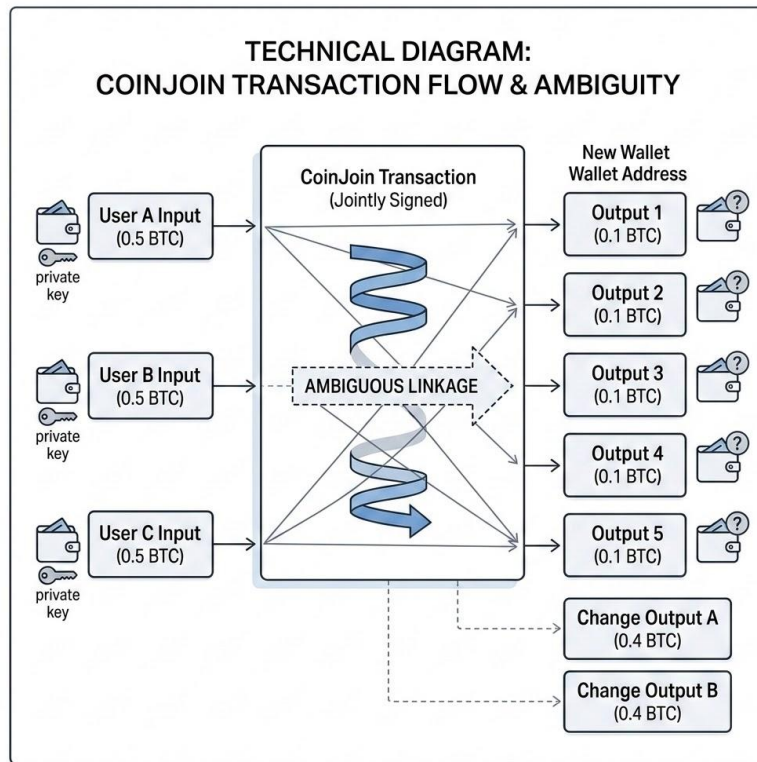


CENTRALIZED CRYPTOCURRENCY MIXER: TRUST & OBSERVABILITY DIAGRAM

- Mixcoin and similar systems use a centralized intermediary to mix coins
- The privacy gain depends on trusting that intermediary not to:
 - log mappings
 - steal funds
 - cooperate with investigators
- Centralized mixers introduce:
 - custody risk
 - legal exposure
 - service-layer surveillance
- They improve privacy operationally, but not structurally
- This makes them useful as temporary tools, but poor as long-term privacy foundations

CoinJoin: Privacy Without Central Custody

- CoinJoin combines multiple users' inputs and outputs into one jointly signed transaction
- Goal:
 - break the simple mapping between sender and recipient
- Unlike centralized mixers, CoinJoin does not require surrendering custody
- It improves privacy by creating ambiguity inside a shared transaction
- Limitations remain:
 - coordination overhead
 - uneven output patterns
 - residual heuristics
- CoinJoin is a significant improvement, but not perfect anonymity



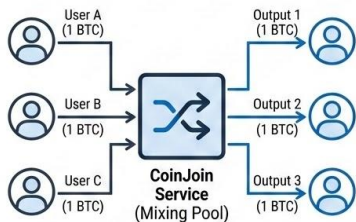
There Are Two Different Strategies for Privacy

PRIVACY APPROACHES IN CRYPTOCURRENCY LEDGERS: WORKAROUNDS VS. DESIGN

PRIVACY BY WORKAROUND (TRANSPARENT LEDGERS)

Privacy features added post-protocol to public, transparent blockchains (e.g., Bitcoin, Ethereum)

Example: CoinJoin



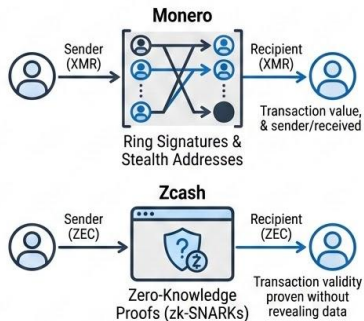
Challenges

-  **Opt-in Only** (Requires Active Participation)
-  **Detectable Patterns** (Heuristic Analysis)
-  **Metadata Leakage** (Timing, IP Addresses)

PRIVACY BY DESIGN (PROTOCOL LEVEL)

Privacy features built into the core protocol, obfuscating transaction data by default

Examples: Monero, Zcash



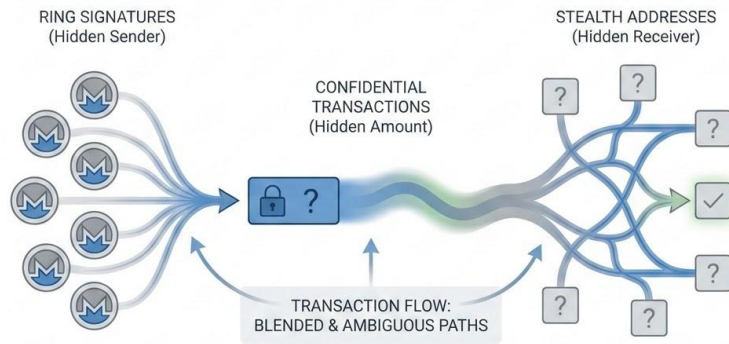
Benefits

-  **Default Privacy** (Mandatory or Strongly Encouraged)
-  **Stronger Obfuscation** (Cryptographic Hiding)
-  **Reduced Metadata Exposure** (Network Level Protection)

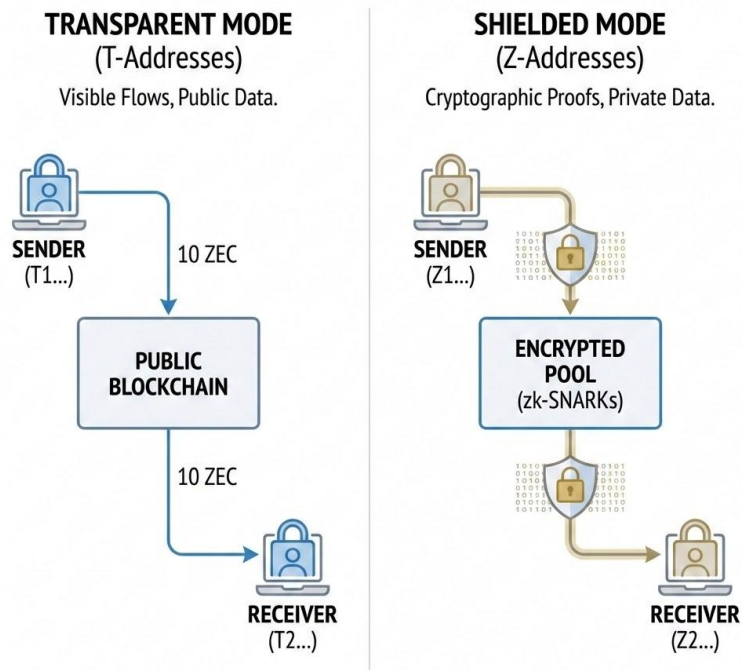
- Strategy 1: start from a transparent ledger and add privacy tools on top
 - Mixers
 - CoinJoin
 - address hygiene
- Strategy 2: redesign the transaction model so privacy is built into the protocol
- The first strategy is operational and fragile
- The second strategy is cryptographic and structural
- This distinction separates Bitcoin privacy workarounds from privacy-first systems such as Monero and Zcash

Monero: Privacy by Default

- Monero was designed so that privacy is not optional, but built in
- It conceals:
 - sender information
 - receiver information
 - transferred amount
- Core mechanisms include:
 - ring signatures
 - stealth addresses
 - RingCT
- The privacy model aims to make every transaction look similar to many others
- This increases the anonymity set and reduces the power of standard blockchain heuristics
- Strong privacy, however, also raises regulatory and exchange-listing challenges



Zcash: Selective Transparency with Zero-Knowledge Proofs

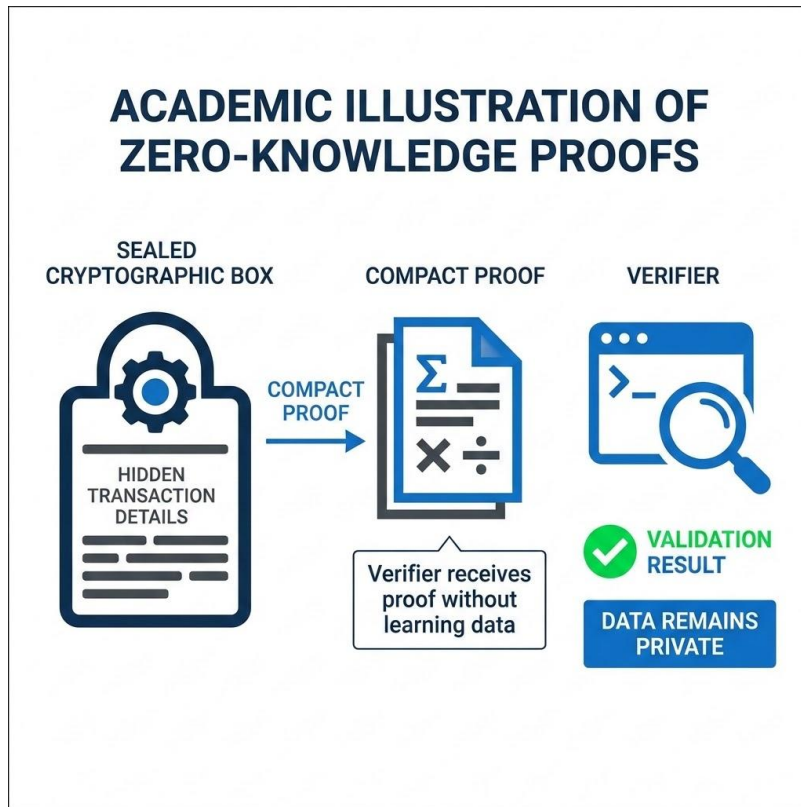


Blue/Gray = Visible Data, Gold Accent = Encrypted/Proven Data

- Zcash takes a different path from Monero:
 - privacy is available, but not always mandatory
- It supports:
 - transparent addresses
 - shielded addresses
- Shielded transactions use zk-SNARKs to prove validity without revealing details
- This model enables a balance between:
 - Confidentiality
 - optional compliance visibility
- Zcash is therefore a key example of privacy designed through proof systems, not only through transaction obfuscation

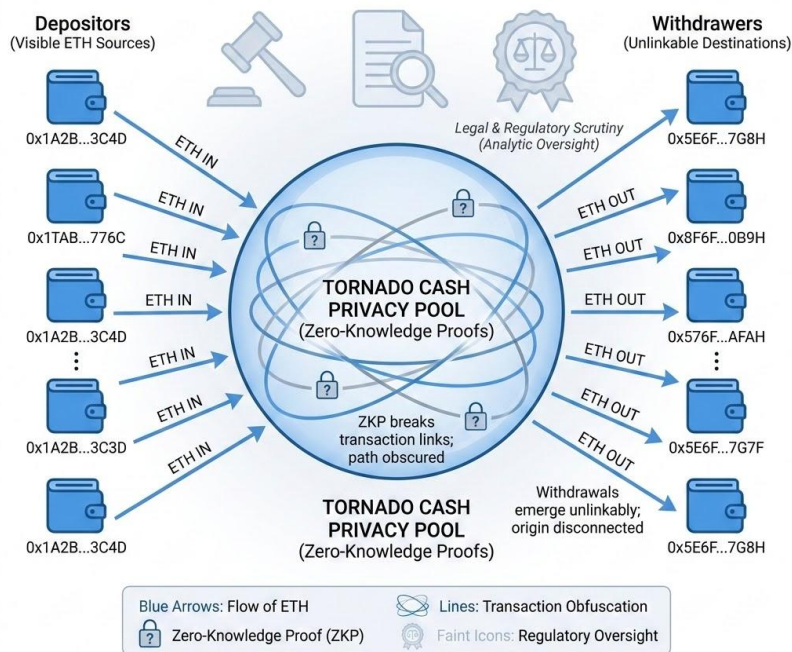
Zero-Knowledge Proofs as a Privacy Primitive

- Zero-knowledge proofs allow a user to prove a statement without revealing the underlying secret
- zk-SNARKs are:
 - Succinct
 - Non-interactive
 - efficient to verify
- They make it possible to prove:
 - transaction validity
 - balance correctness
 - rule compliance
 - without exposing transaction details
- This is a structural shift:
 - from hiding data informally
 - to proving correctness formally
- ZK systems therefore change privacy from an operational practice into a cryptographic primitive



Tornado Cash: Strong Privacy, Strong Backlash

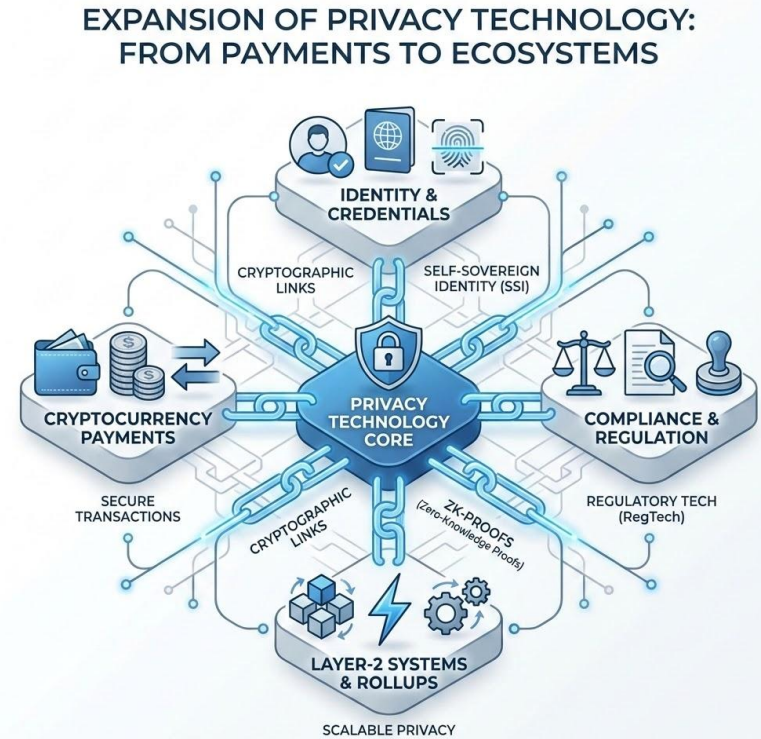
Tornado Cash on Ethereum: Privacy Pool Privacy Pool Mechanism & Regulatory Context



- Tornado Cash applied zk-SNARK-based mixing to Ethereum
- It broke the direct link between deposits and withdrawals through zero-knowledge proofs
- Technically, it demonstrated strong privacy on a transparent smart-contract platform
- Politically and legally, it triggered a major backlash:
 - Sanctions
 - Arrests
 - global debate about code, responsibility, and laundering
- Tornado Cash is therefore not only a technical case, but a governance and legal turning point for privacy tools

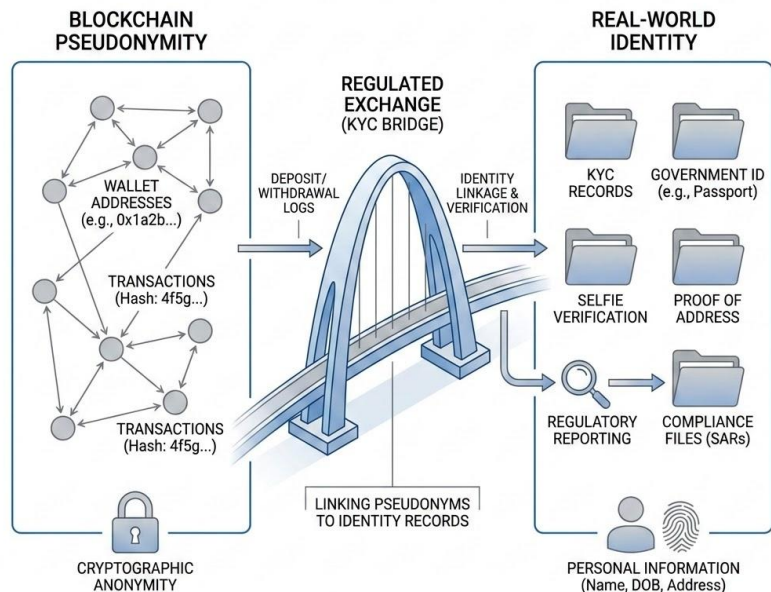
Privacy Is Expanding Beyond Payments

- Privacy-preserving cryptography is no longer limited to private payments
- New directions include:
 - Zk-rollups
 - privacy-preserving compliance
 - selective disclosure
 - digital identity proofs
- Layer-2 systems use proofs to compress computation and reduce on-chain exposure
- The same cryptographic logic that hides value can also hide personal data
- This is where blockchain privacy converges with digital identity and verifiable credentials



Why Exchanges Undo So Much Privacy

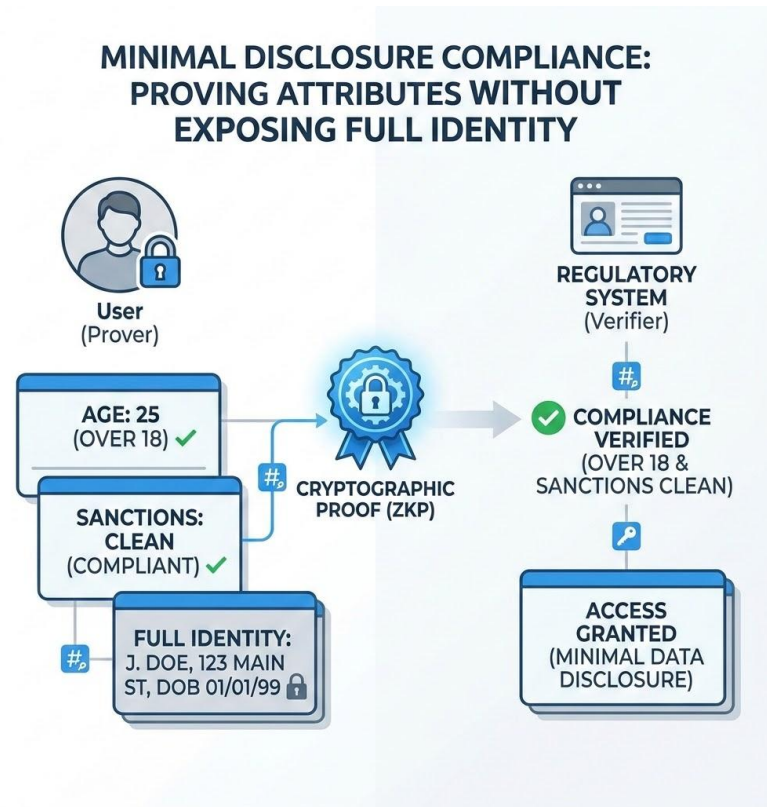
BLOCKCHAIN DEANONYMIZATION THROUGH KYC



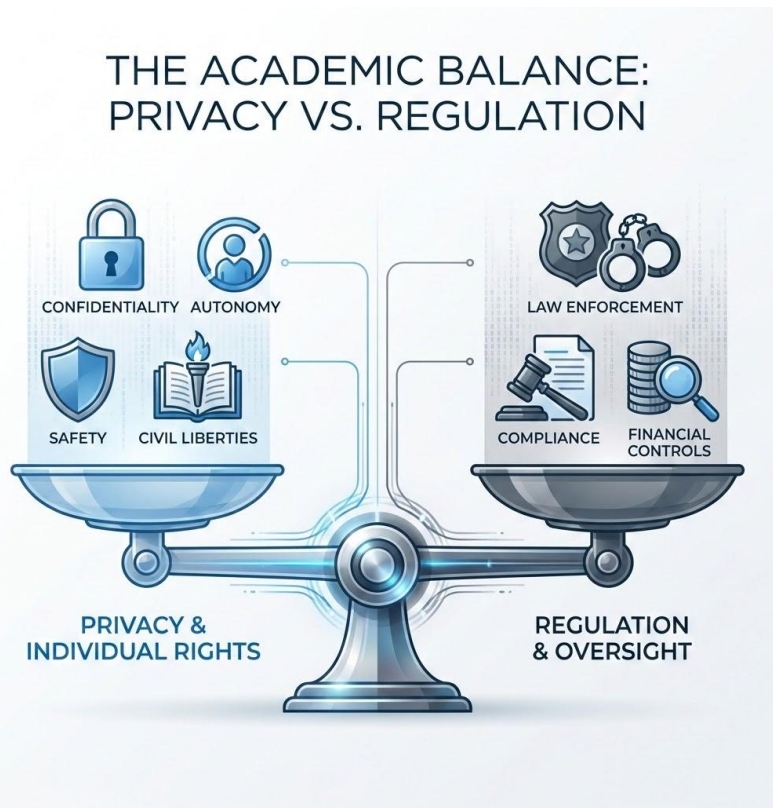
- Public blockchains may be pseudonymous, but exchanges often are not
- KYC ties wallets to:
 - legal identity
 - national identifiers
 - Photographs
 - institutional records
- Once funds cross a compliant exchange boundary, on-chain activity becomes easier to associate with real people
- This creates a structural bridge between public transaction graphs and state-recognized identity
- As a result, privacy in cryptocurrency is often weakest where the blockchain meets the regulated financial system

Can We Have Compliance Without Full Disclosure?

- A growing research direction is privacy-preserving compliance
- Instead of revealing all personal information, a user proves only what is required
- Examples:
 - over-18 proof
 - sanctions compliance
 - solvency proof
 - regulated identity eligibility
- This approach combines:
 - zero-knowledge proofs
 - selective disclosure
 - verifiable credentials
- The goal is not privacy instead of regulation, but compliance with minimal disclosure



The Privacy Debate Is Also a Governance Debate

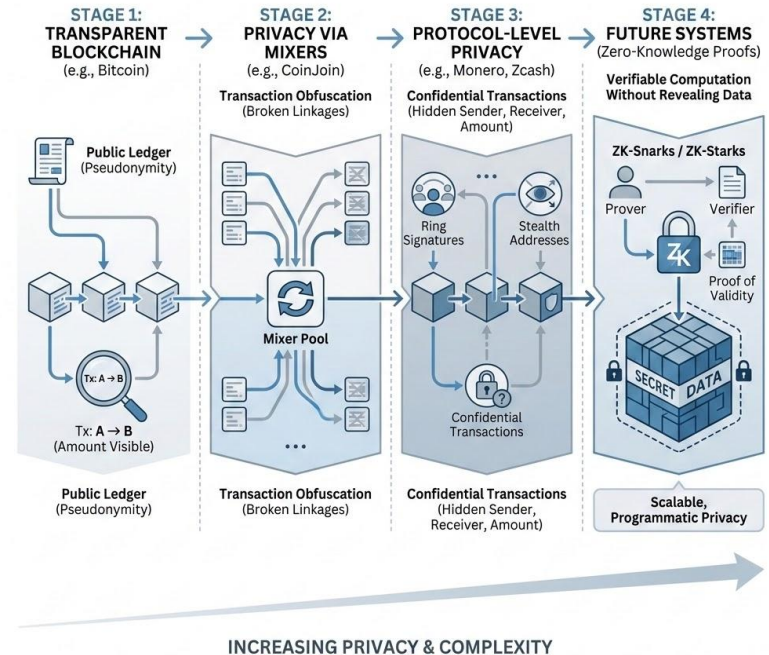


- Strong privacy protects:
 - political freedom
 - personal safety
 - Autonomy
 - confidentiality
- Strong privacy can also complicate:
 - law enforcement
 - financial controls
 - sanctions enforcement
- Backdoors and exceptional access create their own systemic risks
- The real policy question is not whether privacy has value, but how much surveillance a society is willing to normalize
- Cryptographic design therefore becomes a political design choice

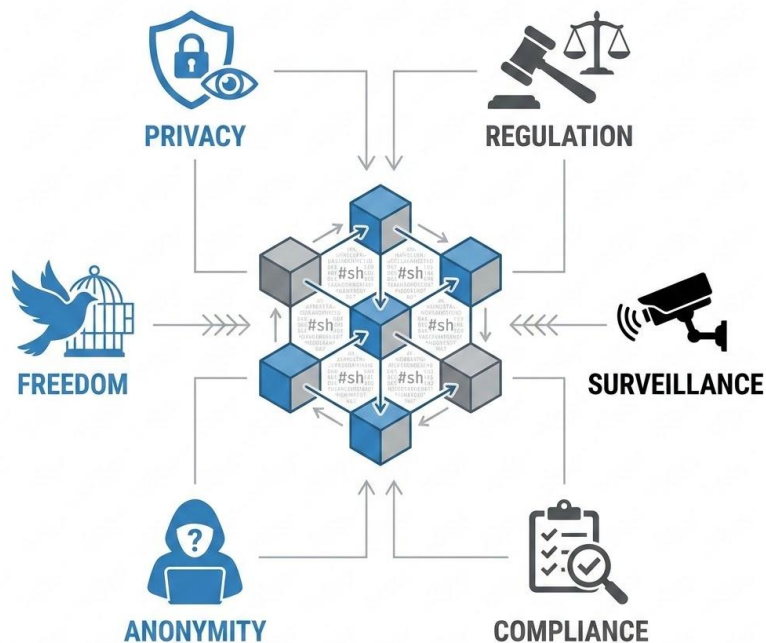
Summary and Takeaways

- Bitcoin is not truly anonymous; it is public, traceable, and behaviorally linkable
- Privacy in blockchain fails because transparency is built into the ledger structure
- Mixers and CoinJoin improve ambiguity, but do not fully solve the structural problem
- Monero and Zcash show two different privacy-by-design responses
- Zero-knowledge proofs are changing privacy from workaround to protocol primitive
- The next generation of privacy systems will likely combine:
 - Cryptography
 - Compliance
 - selective disclosure
 - identity-aware proofs

EVOLUTION OF BLOCKCHAIN PRIVACY MECHANISMS: A SUMMARY



Discussion and Reflection



- Is privacy in finance a right, a risk, or both?
- Should privacy be:
 - mandatory by default
 - optional by user choice
 - limited by regulation?
- How should governments respond to privacy-preserving code?
- Can zero-knowledge systems realistically satisfy both confidentiality and accountability?
- Which is more dangerous:
 - mass surveillance
 - or unbounded secrecy?

Closing Idea

- Public ledgers do not become private just because names disappear
- Privacy must be engineered deliberately:
 - Structurally
 - Cryptographically
 - institutionally
- The future of blockchain privacy will be decided not only by code, but by the conflict between transparency, control, and freedom

